#HW 3

Graded

Student

Ingrid Villagomez

Total Points

94 / 100 pts

Question 1

Q1

10 / 10 pts

- ✔ **– 0 pts** Correct
- **– 3 pts** Your calculation and graph is not clear

Question 2

Q2

10 / 10 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** There is a mistake in calculation
- **– 4 pts** Incomplete

Question 3

Q3

10 / 10 pts

- ✔ **– 0 pts** Correct
- **– 4 pts** not clear
- **– 5 pts** incomplete

Question 4

Q4

8 / 10 pts

- **– 0 pts** Correct
- ✔ **– 2 pts** error in calcuation
- **– 5 pts** incomplete

Question 5

Q5

10 / 14 pts

– 0 pts Correct

– 2 pts Final pressure is wrong.

✔ **– 4 pts** unclear

– 4 pts incomplete

– 2 pts unit conversion not done.

Question 6

Q6

46 / 46 pts

✔ **– 0 pts** Correct

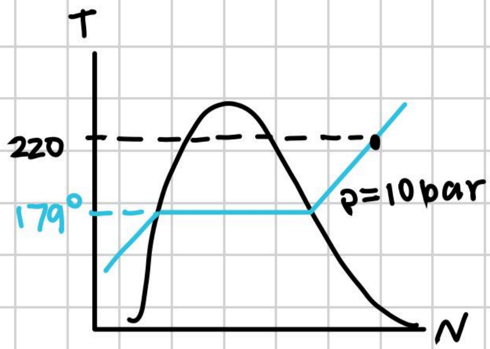
– 0 pts Click here to replace this description.

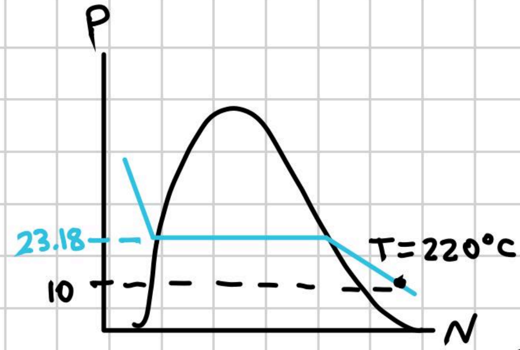
No questions assigned to the following page.

Questions assigned to the following page: 1, 2, and 3

1) $P = 10\ bar$

$T = 220°C$





Superheated vapor: $v = 0.21675\ m^3/kg$

$u = 2657.4\ \frac{kJ}{kg}$

2) $P = 20\ bar$

$u = 300\ kJ/kg$

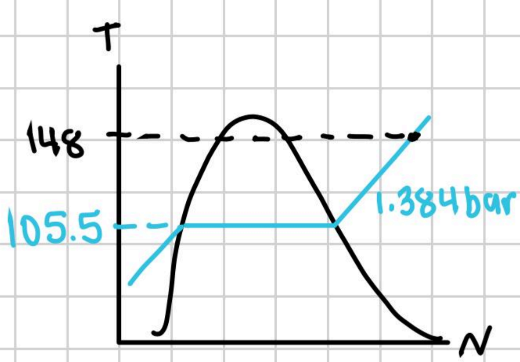
$\frac{m_{vap}}{m_{liq.} + m_{vap}} = x$

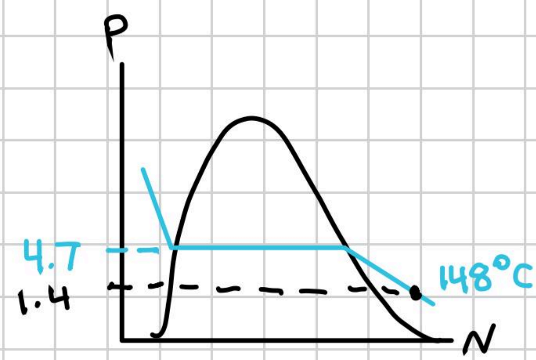
$$x = \frac{m_{vapor}}{m_{total}} = \frac{u - u_f}{u_{fg} - u_f} = \frac{300 - 251.3}{478.1 - 251.3} = 0.214$$

3) a) $T = 300\ °F$ $P = 20\ lbf/in^2$

$T = 148.9°C$ $P = 1.384\ bar$

$\frac{lbf/in^2}{14.455}$



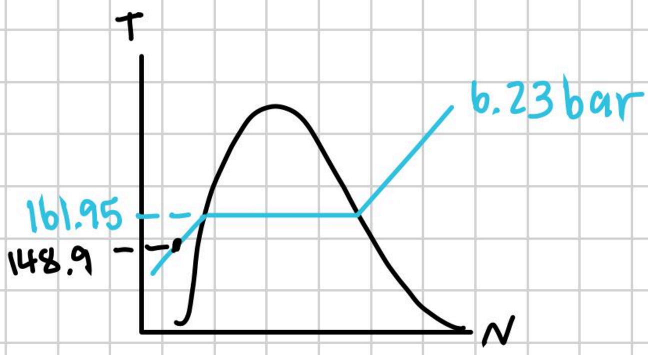


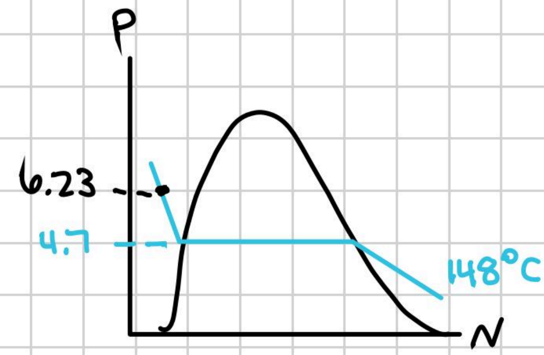
Superheated vapor

b) $T = 148.9°C$ $P = 90\ lbf/in^2$

$P = 6.23\ bar$

Questions assigned to the following page: 3 and 4

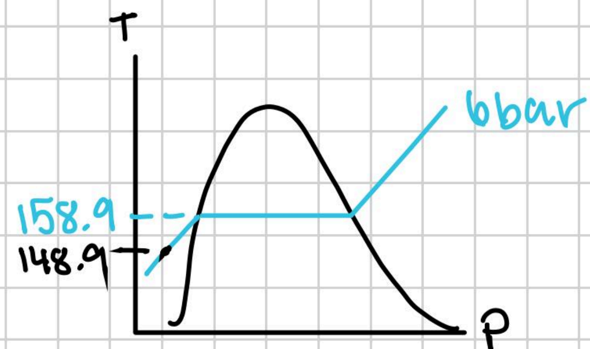


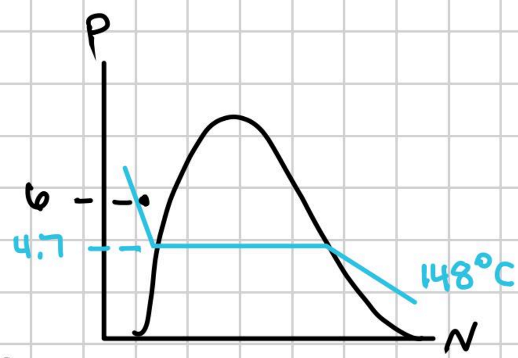


compressed liquid

c) $T = 148.9°C$ $\quad v = 5\,ft^3/lb \times 0.062428 \frac{m^3}{kg}$

$= 0.3121 \frac{m^3}{kg} \rightarrow P = 6bar$





compressed liquid

4) $m = 2lb$ $\quad V = ?$ $\quad P = ?$ $\quad v_f = 0.01611\,ft^3/lb$

$T = 40°F = 4.44°C$ $\quad v_g = 0.0981\,ft^3/lb$

Quality $x = .20$

$$v = v_f + x(v_g - v_f)$$

$$v = 0.01611 + (0.20)(0.0981 - 0.01611)$$

$$v = 0.03251\,ft^3/lb$$

$$V = m \times v = 2lb \times 0.03251\,ft^3/lb = 0.06502\,ft^3$$

$$P = P_1 + \frac{T - T_1}{T_2 - T_1}(P_2 - P_1)$$

15.12 table zybooks

Questions assigned to the following page: 4, 5, and 6

$$P = 3.2 + \frac{(4.44-2.48)}{(8.93-2.48)}(4-3.2) = 3.44\,\text{bar}$$

$x = 0.3$

5) $T = 93.3°C \longrightarrow T_2 = 115.6°C$

$P_i = 0.77345\,\text{bar} \quad P_f = 1.709\,\text{bar}$

$P_i = 0.6535\ \text{lbf/in}^2 \quad P_f = 0.1182\ \text{lbf/in}^2$

$v_f = 1.038 \times 10^{-3}\ \text{m}^3/\text{kg} \quad v_{f_2} = 1.05595 \times 10^{-3}\ \text{m}^3/\text{kg}$

$v_g = 2.172\ \text{m}^3/\text{kg} \quad v_{g_2} = 1.05095\ \text{m}^3/\text{kg}$

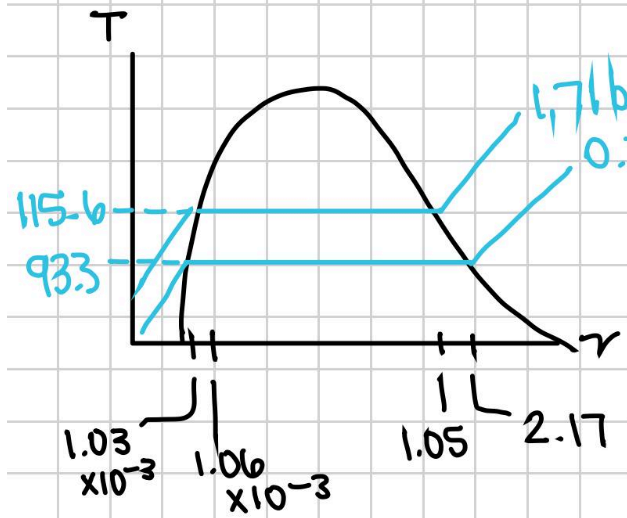
$$v = v_f + x(v_g - v_f)$$

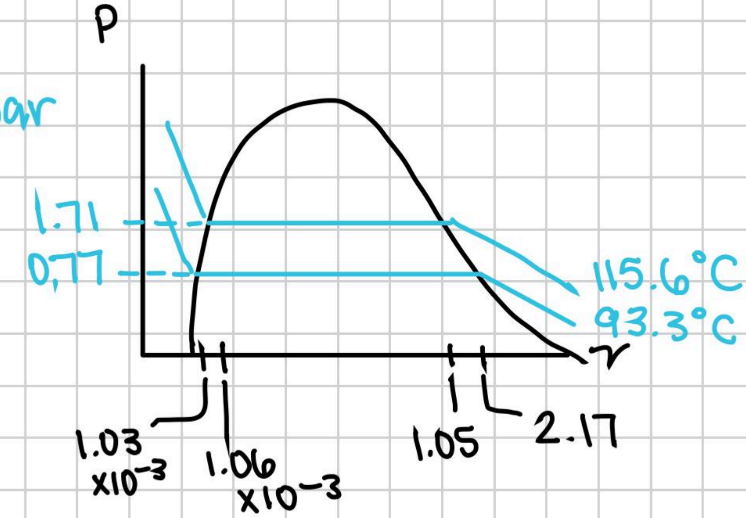
$$= 0.0167 + (0.3)(34.79 - 0.0167)$$

$$= 10.4487\ \text{ft}^3/\text{lb}$$

$0 < x < 1$

S.h.m.





liquid vapor mixture